Conference Paper Title*

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Sakshi Kurian
Computer Engineering
Fr. C.R.I.T
Mumbai, India
@kuriansakshigmail.com

Janhvi Suryavanshi
Computer Engineering
Fr. C.R.I.T
Mumbai, India
jnsuryavanshi14@gmail.com

Arnold George
Computer Engineering
Fr. C.R.I.T
Mumbai, India
@arnoldgeorge0016gmail.com

Canute Mendonza
Computer Engineering
Fr. C.R.I.T
Mumbai, India
@gmail.com

***Abstract*—Campus life presents students with a variety of daily challenges, including navigating in campus, accessing academic resources, having official platform for community discussion and managing canteen services. These fragmented systems often lead to inefficiencies, increased time consumption, and a lack of seamless access to essential services.To address these issues, CampusHubX is introduced as an integrated and intelligent digital platform that consolidates resources sharing and discussion, digital canteen coupon and digital navigation system into a unified, user-friendly ecosystem. Developed using robust backend technologies such as PHP and supported by a centralized MySQL database, the system ensures efficient data processing, real-time updates, and secure transactions.By centralizing key campus functionalities, CampusHubX significantly reduces operational complexity, enhances user convenience, and fosters collaboration among users. The platform also promotes sustainability by encouraging the reuse of academic resources and minimizing reliance on paper-based processes. Ultimately, CampusHubX contributes to the digital transformation of educational institutions by streamlining campus operations, improving accessibility, and delivering a smarter, more connected campus experience.**

***Index Terms*—Keywords: CampusHubX, Canteen Coupon, Resource Sharing and Discussion, Digital Navigation, Web Technologies,Dijkstra Algorithm, Sustainable Systems, User Convenience**

I. Introduction

A. Background

Not just student but also staff or any other people in college often struggle with various problems like canteen queue, validated resources, finding a classroom in their college life which leads to underutilization of their time and effort. As a result, the effort and time they could utilize for working is being wasted in this struggle. Therefore , building a platform that will address these problems by integrating the services at one place. It provides user to access the resources , having a discussion platform , ordering canteen food with token system also navigating through campus using dijskastra algorithm.

B. Motivation

The main motivation is to improve the life of campus user by saving time and bringing the necessary facilities to an integrated platform.Including providing quick access to campus locations using Dijskstra algorithm. In addition to it encouraging resources sharing and building a community to discuss various topics or solving queries.Reducing the long queues creating chaos in the canteen by building safe transaction based Canteen Coupon generation.Overall, to enhance the student experience with a single connected system.

C. Aim and Objective

The main objective is to build an integrated platform that allows campus people to easily communicate with each other, sharing, and accessing validated resources. Reducing the queues in the canteen by generating an online coupon through which the user can purchase the order. By using Dijkstra's algorithm, navigating the destined path. Thus, our project comprises the issues and addresses it at one place.

II. Ease of Use

The reviewed literature highlights the growing role of digital platforms and smart technologies in improving efficiency, accessibility, and user experience across various domains. Navale et al. [1] demonstrated how localized navigation systems can enhance movement within college campuses using structured mapping and guidance systems. Salunke and More [2] emphasized the importance of peer-to-peer platforms, showing how digital systems can facilitate resource sharing like book exchange while promoting collaboration among users.

Rajath Bagre et al. [3] introduced RFID-based smart canteen systems, illustrating how automation can streamline operations, reduce waiting time, and improve transaction accuracy. Verma and Varshney [4] explored smart campus models using MANET and optimization algorithms, highlighting the integration of sensors and intelligent networking to create efficient and adaptive environments.

Jiang et al. [5] and Xing et al. [6] focused on digital coupon systems, explaining how small-value incentives and platform strategies influence consumer behavior and spending patterns. These studies underline the importance of user engagement and economic incentives in digital platforms.

Luo and Chen [7] presented a private book-sharing system based on the sharing economy model, demonstrating how technology can support sustainable resource utilization and community-based services.

Overall, these studies collectively suggest that integrating technologies such as IoT, RFID, digital platforms, and intelligent algorithms can significantly improve system efficiency, transparency, and user satisfaction. They also highlight the importance of decentralized systems, user interaction, and data-driven decision-making in developing modern smart applications.

III. Proposed System

A. Problem Statement

In most college campuses, users face difficulty accessing essential campus facility. They often struggle to locate classrooms or departments, finding relevant resources and solving query facilites, waiting for long queues in canteen. Currently, these services are handled manually or through separate systems,leading to inconvenience, time wastage, and communication gaps. CampusHubX aims to solve these problems by integrating multiple features such as Digital Navigation System, Resource Sharing and Discussion and Digital Coupon System into a single unified platform. This system provides path based campus navigation, simplifies online resource sharing among students, and QR based canteen coupon generation process to make campus life smoother and more organized.

B. System Overview

The proposed system would consist three modules i.e Resource Sharing and Discussion, Digital Canteen Coupon and Digital Navigation System. Our website will be used by three categories user, admin and canteen admin.

Phase 1: Admin Registration The system is based on a centralized single-admin model, where the administrator has complete control over user management, system administration, and content regulation. The admin is responsible for admitting users into the system and can create user accounts individually or in bulk by uploading a CSV file. Through this, the efficiency of enrollment is increased. Along with it the administrator also keeps track of issues reported by users within the platform. When a user reports a problem regarding any content posted, it will be reviewed by the administrator. The administrator, upon evaluating the content, will make a decision on whether to keep it or remove it from the platform. This method ensures proper content regulation, maintains system integrity, and improves the overall user experience.

Phase 2: User Registration In this phase the system lets new users sign up by filling out a form with their details. The form has checks to ensure all the information is correct and complete.When the user submits the form their account is not activated away. It stays pending until an administrator reviews. Approves it. During this time the user can't use the system.After the account is approved the user can log in. Start using the platform. When they log in they will see three modules and can choose any of them based on what they want to do. This makes the system easy and flexible to use for the user registration process. The user can access the modules after the user registration is complete.

Phase 3: Resource Sharing and Discussion This module consist of resource sharing and discussion which helps users share materials and talk about content on the platform. Starting with resource sharing ,here users can upload up to five files at a time with multiple formats like XML, JPG, JPEG, Excel, CSV, PPT, DOC, and PDF. There is no need to explicitly mention the title, it is automatically taken from the file name. Users can add a description to give details about the file. After the file is uploaded it is accesible for other where they can view and download it. Files are automatically deleted after 30 days to save storage space. On the other part the discussion section lets users talk about various topics. They can post a query/information along with any attachment. Once posted the other users can join the conversation and express their thought by commenting or replying. Posts are filterized into groups like SOS, General, Coding, Lost and Found. Each group is represented with a color, so it is easy to differentiate them. SOS posts are given highest priority by showing at top. Users can report posts, comments, or files that are not suitable. The administrator looks at these reports and takes action to keep the platform fair and positive. This makes sure the platform is helpful and trustworthy for everyone. The Resource Sharing and Discussion module helps users share knowledge and stay informed.

Phase 4: Digital Canteen Coupon This module helps user order food digitally. They get a generated coupon pdf. The process is not only easy but also saves time making the system more organized.After entering this module user will see the food items on the dashboard.The food items are grouped by categories like breakfast, snacks, drinks, etc. If any item is not available in the canteen, the food card will be disabled. Users can pick any item ,choose the quantity and add it to their cart. After this are redirected to the payment page where they will be able to make payment by scanning QR code and will upload the screenshot after which they will submit it. The canteen owner will review the payment and will admit it. After the order is accepted the user will get notified as well as will get the digital coupon. The coupon has details like the order number, ordered food, its quantity, total cost, date and time. Each coupon also has token number which gets updated on daily basis. This helps manage orders easily. Users can download the coupon and will show it at the canteen counter when they pick up their food. The staff checks the coupon on the system. When they get their food the coupon is discarded. The system also keeps a record of all orders. Users can see their orders. They can check the status like if it was approved. They can reorder any previous order also can add more items to it. Overall this module helps in saving time of the campus users by reducing the long queues for ordering.

Phase 5: Digital Navigation System This module has a navigation feature that helps people who use the campus navigation feature find places on campus quickly. People who use the campus navigation feature start by saying where they are now and then they choose where they want to go, like a classroom or the canteen. The campus navigation feature system then works out the route. The route is shown on the screen with a map and step by step directions. The directions are easy to follow. The campus navigation feature system can also give voice instructions, which is helpful for people who use the campus navigation feature and do not want to look at the screen. The campus navigation feature is designed to be easy to use and accurate. It helps people who use the campus navigation feature reduce confusion and unnecessary walking around the campus. The campus navigation feature is especially useful for students, visitors or anyone who does not know the campus well. This module makes it easier for people who use the campus navigation feature to get there, it saves time and makes the campus navigation feature more comfortable. The campus navigation feature helps people who use the campus navigation feature find their way around the campus. It provides an experience, for all people who use the campus navigation feature.

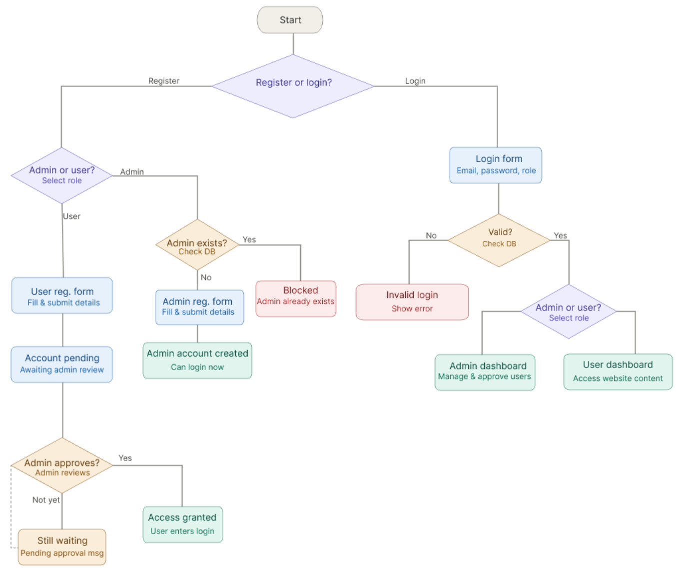
C. Key Features and Functionalities

The proposed CampusHubX system provides the following key features:

- **Multi-File Upload Support:** The system allows users to upload multiple files simultaneously, supporting various file formats such as PDF, images, documents, and presentations, ensuring flexibility in resource sharing.
- **Automatic File Cleanup Mechanism:** The platform includes an automated cleanup feature that removes files older than 30 days, optimizing storage usage and maintaining system efficiency.
- **Automated Inventory-Based Menu System:** The canteen module dynamically updates menu availability based on inventory, ensuring users can only order items that are currently in stock.
- **Directory-Based File Management:** The system utilizes a structured directory-based path system for efficient file storage, retrieval, and organization, improving performance and maintainability.

IV. System Diagrams

Fig 1. shows the registration and login process.Admin has to register by filling the details and after that no more admin will be generated. The user will register and wait for admin approval to access the website. After the approval the user will be able to access the website Fig 2. shows the flow of the system where after logging the user will have 3 modules to select depend on the work. After selecting the resource sharing and discussion, either you can post a query or share any material. After posting it will be open to all for accessing it. In the digital canteen coupon, the user will add the food from menu, which will be loaded to cart. It will be navigated to payment page where by scanning the QR the payment is to be done. After completing the transaction, screenshot must be uploaded and submitted. The coupon will only get generated after the approval of owner until the status will remain pending. Whereas, in the digital navigation section, the user will select the source and destination.Path will get generated step by step with references of nearby classes. In addition, voice command will also be provided where the steps will be readed. Fig 3. highlights the communication between the canteen owner side and user side through the platform. The owner will update the menu by toggling it off or on depending



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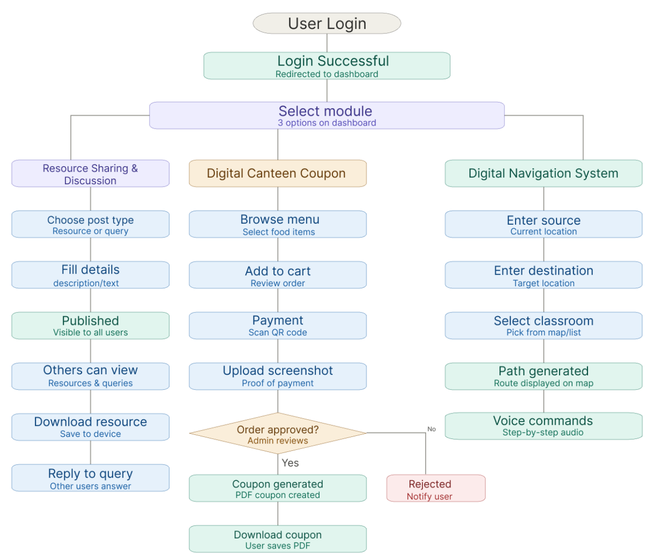

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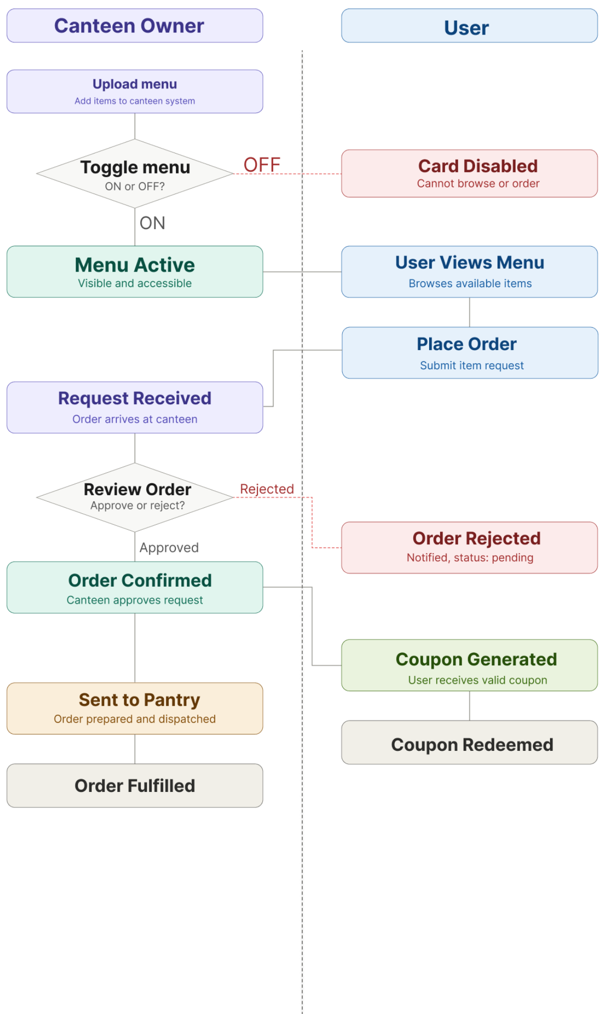

Fig. 3. Enter Caption

on the availability of the item. The user can see all the items, but can only access items which are available. After the order is placed it will be added in the queue in pending mode until the owner confirms by reviewing it. After the canteen admin accepts it, along with the coupon generated on user side the order will go to pantry.

V. Conclusion

In conclusion, CampusHubX provides a centralized platform making campus life easier for users. The system ensures security through verification based user registration. It follows single administration apporoach which maintains system integrity, content regulation.The future scope of this system is vast, with potential for further improvements and innovations.One area of the future development could be the expansion of the system to use payment gateway. Additionally, the system could be expanded to include whole campus for navigation.Overall, the proposed CampusHubX system has the potential to significantly enhance campus efficiency by offering a secure, integrated, and user-centric platform for resource sharing, communication, navigation and service management.

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